

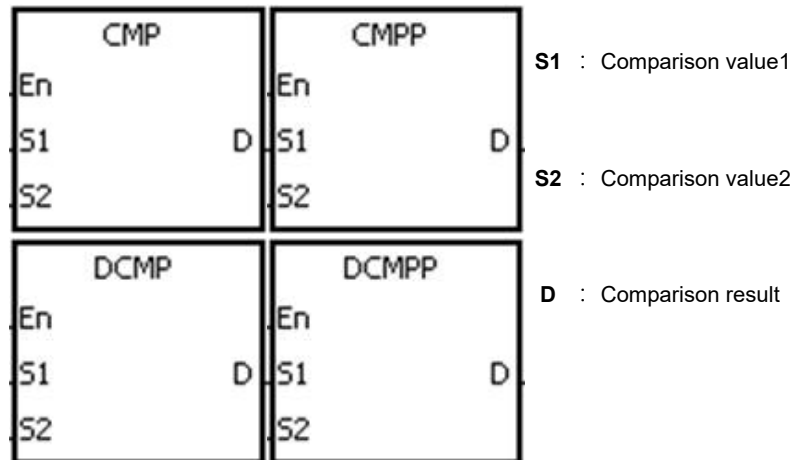
API	Instruction code			Operand	Function
0054	D	CMP	P	$S_1 \cdot S_2 \cdot D$	Comparing values

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1	●	●			●	●	●	●	●		○	○	○	○		
S_2	●	●			●	●	●	●	●		○	○	○	○		
D		●	●	●				●								

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1		●	●			●	●				●	●	
S_2		●	●			●	●				●	●	
D	●												

Pulse Instruction	16-bit instruction	32-bit instruction
AS	AS	AS

Symbol

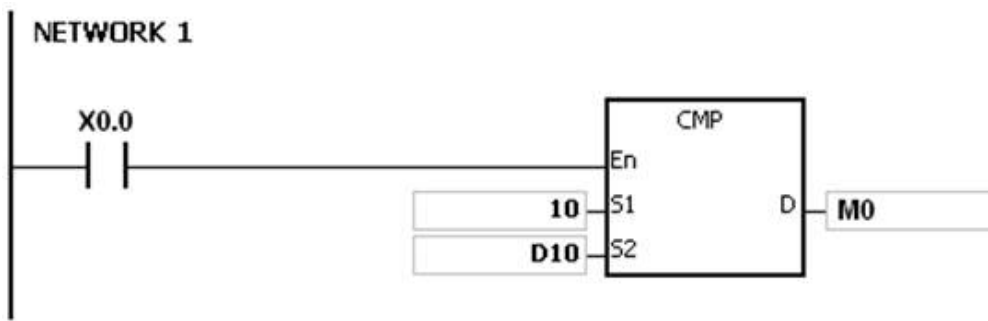


Explanation

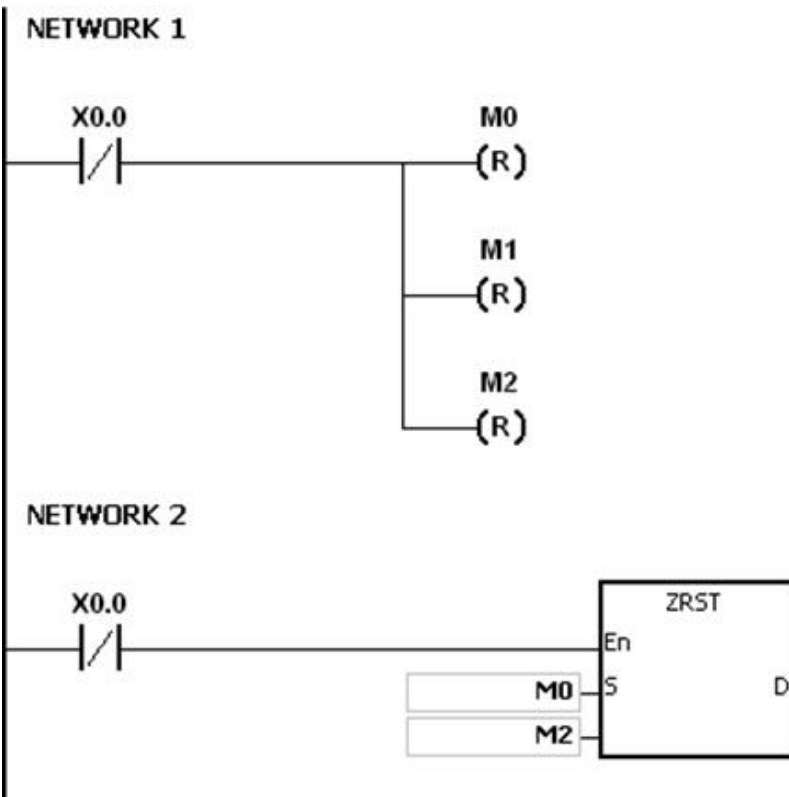
- These instructions compare the single decimal numbers in S_1 and S_2 and store the comparison results in D.
- The operand D occupies 3 consecutive devices. The comparison results are stored in D, D+1, and D+2. If the value in S_1 is greater than the value in S_2 , D is ON. If the value in S_1 is equal to the value in S_2 , D+1 is ON. If the value in S_1 is less than the value in S_2 , D+2 is ON.
- Only the DCMP and DCMPP instructions can use the 32-bit counter, but not the device E.

Example

- If the operand D is M0, the comparison results are stored in M0, M1 and M2, as shown below.
- When X0.0 is ON, the CMP instruction is executed. M0, M1, or M2 is ON. When X0.0 is OFF, the execution of the CMP instruction stops and the state of M0, the state of M1, and the state of M2 remain unchanged.



3. If you need to clear the comparison result, use the RST or ZRST instruction.



Additional remarks

1. If you declare the operand **D** in ISPSOft, the data type is ARRAY [3] of BOOL.
2. If **D+2** exceeds the device range, the instruction is not executed, SM0 is ON, and the error code in SR0 is 16#2003.